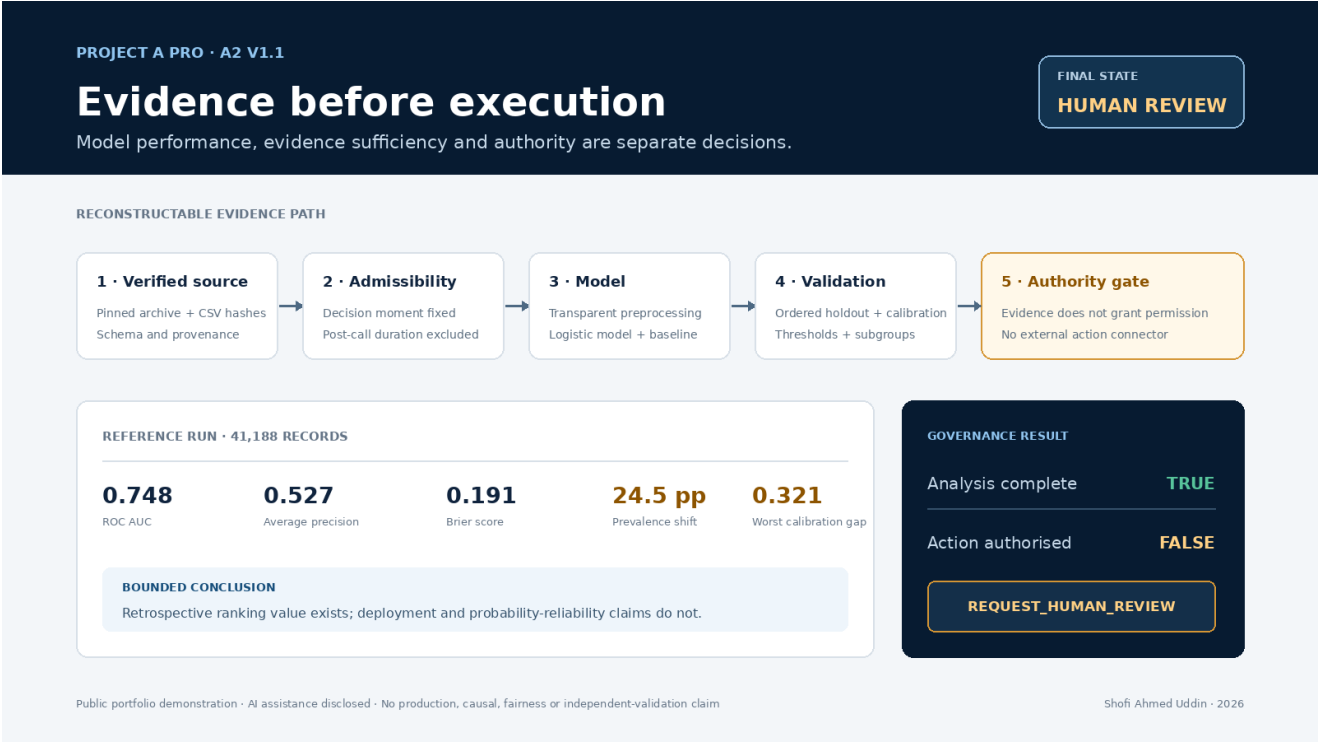


Evidence before execution

A working Data Science system that separates model performance, evidence sufficiency and authority to act.



FINAL DISPOSITION	ANALYSIS COMPLETE	ACTION AUTHORISED
REQUEST_HUMAN_REVIEW	TRUE	FALSE

What this proves

A2 did not merely produce a predictive model. It detected where apparently useful statistical evidence stopped justifying execution. The system excluded post-contact leakage, exposed severe temporal shift and high-band miscalibration, and kept the final decision at accountable human review.

The evidence and the boundary

A reproducible technical demonstration using 41,188 pinned historical campaign records from the UCI Bank Marketing dataset.

Measure	Result	What it supports
ROC AUC	0.748	Bounded retrospective ranking value
Average precision	0.527	Minority-class discrimination
Brier score	0.191 vs 0.273	Improvement over prior baseline
Calibration ECE	0.093	Material probability error remains
Worst bin gap	0.321	High-score overconfidence
Train / holdout prevalence	6.4% / 30.8%	Severe temporal shift; no deployment claim

Controls demonstrated

PROVENANCE	Pinned source and analytical-file hashes; schema and target validation.
ADMISSIBILITY	Call duration removed because it is unavailable at the pre-contact decision moment.
HARDENING	Calibration, threshold and subgroup diagnostics plus hostile target-copy and corrupted-label tests.
AUTHORITY	Analytical completion is distinct from permission. Critical evidence failure remains non-authorising even when approval is supplied.

Commercial application

The control pattern can be applied to one bounded workflow in financial services, fraud operations, insurance, health, public services, safety assurance or autonomous-system review. A client engagement would reconstruct the decision moment, permissible evidence, uncertainty, authority boundary, safe-stop conditions and replay record before production integration is considered.

This is not a deployable banking product, causal study, fairness audit, security certification, legal opinion, independent validation or autonomous decision engine.

Dataset: S. Moro, P. Rita and P. Cortez, Bank Marketing, UCI Machine Learning Repository, DOI 10.24432/C5K306, CC BY 4.0. Code: MIT. Material generative-AI assistance is disclosed; Shofi Ahmed Uddin remains responsible for architecture, statistical judgement, claims and publication.